

ESP32-S3 Overview

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1. What Is the ESP32-S3

The ESP32-S3 is an **AIoT SoC** released by Espressif Systems in 2021. Built on the classic ESP32, it significantly strengthens the **AI acceleration capability**, **IO resources**, and **native USB support**, targeting scenarios such as screen-equipped smart devices, voice interaction, and image recognition.

Its core positioning is:

Wi-Fi + Bluetooth 5 (LE) + AI vector accelerator + rich GPIO

2. Core Chip Specifications

Parameter	Specification
CPU	Xtensa 32-bit LX7 dual-core, up to 240 MHz
AI accelerator	Vector extension instructions, supports 8/16/32-bit integer operations
SRAM	512 KB (internal)
ROM	384 KB (fixed Bootloader, low-level libraries)
Wi-Fi	802.11 b/g/n (2.4 GHz), HT20/HT40, up to 150 Mbps
Bluetooth	BLE 5.0 + Bluetooth Mesh
USB	Native USB OTG 1.1 (supports Device and Host modes)
GPIO	45 programmable GPIOs (some multiplexed for analog functions)
ADC	2 × 12-bit SAR ADC, up to 20 channels
Touch	14 touch sensors (capacitive)
SPI	4 units (2 configurable as Octal PSRAM; pins can be remapped to any GPIO)

Parameter	Specification
I2C	2 units
I2S	2 units (supports TDM multi-slot mode)
UART	3 units
TWAI	1 unit (CAN 2.0 compatible)
RMT	8 channels (IR / WS2812 / DShot, arbitrary waveforms)
LED_PWM	8 channels of independent PWM
MCPWM	2 units (motor control PWM, 6 outputs)
USB Serial/JTAG	Built-in USB-to-Serial/JTAG controller; flash and debug without an external UART chip
LCD	Supports 8080 / RGB / SPI / I2C interfaces
Camera	Supports DVP 8/16-bit parallel camera interface
SD/MMC	2 SD Host units (3.3V), up to 40 MHz
Security	Secure Boot, Flash Encryption, AES/SHA/RSA/HMAC, Digital Signature
Package	QFN56 (7 × 7 mm)
Operating temperature	−40°C ~ +85°C

3. Comparison with Classic ESP32

Feature	ESP32	ESP32-S3
CPU cores	Xtensa LX6 dual-core	Xtensa LX7 dual-core
Max frequency	240 MHz	240 MHz
AI vector acceleration	None	Yes (PIE + SIMD vector coprocessor)
SRAM	520 KB	512 KB
Bluetooth	BLE 4.2 + Classic	BLE 5.0 (no Classic)
USB OTG	None (requires external PHY)	Native USB OTG 1.1
USB Serial/JTAG	None	Built-in, no UART needed for flashing
GPIO	34	45
SPI	4 units	4 units (2 can be Octal)
Touch sensors	10	14

Feature	ESP32	ESP32-S3
RMT	8 channels	8 channels
LCD interface	None	8080 / RGB / SPI / I2C
Camera interface	None	DVP 8/16-bit

The ESP32-S3 drops Bluetooth Classic (BR/EDR) and only keeps BLE. If you need Classic Bluetooth (A2DP audio, etc.), choose the ESP32 instead.

4. Modules and Variants

The ESP32-S3 chip is usually sold as **modules**. A module integrates the chip + Flash + PSRAM + antenna matching circuit, and is soldered directly onto the development board PCB, so users do not need to handle RF layout.

The module used in this tutorial is:

Official Espressif Module: ESP32-S3-WROOM-1-N16R8

Designator	Meaning
ESP32-S3	Chip name
WROOM-1	Built-in PCB antenna, dimensions 18.0 × 25.5 mm
N	Flash package type: N = 8-line Octal SPI (optional Q = Quad SPI)
16	Flash capacity: 16 MB
R	With PSRAM (Pseudo-Static RAM)
8	PSRAM capacity: 8 MB (Octal SPI)

16 MB Flash + 8 MB Octal PSRAM is a high-end configuration among mass-produced modules, easily capable of running the LVGL graphical interface, the TinyUSB protocol stack, and complex AI inference models.

Other common variants:

Module model	Flash	PSRAM	Suitable for
ESP32-S3-WROOM-1-N4	4 MB	None	Lightweight Wi-Fi sensors
ESP32-S3-WROOM-1-N8R2	8 MB	2 MB (Quad)	Simple GUI / audio streaming
ESP32-S3-WROOM-1-N16R8	16 MB	8 MB (Octal)	AI + voice + GUI full scenarios

Module model	Flash	PSRAM	Suitable for
ESP32-S3-WROOM-1U-N16R8	16 MB	8 MB (Octal)	Same as above + external antenna (IPEX connector)

5. Target Application Scenarios

Field	Typical application
AI voice	Offline speech recognition (WakeNet / MultiNet), XiaoZhi AI assistant, smart speakers
HMI display	Smart panels, 86-box central control screens, cycling computers, thermostat LCD screens
Image recognition	Face detection, gesture recognition, QR code scanning (requires external DVP camera)
IoT gateway	MQTT gateway, Zigbee-to-Wi-Fi bridge, Modbus TCP/RTU conversion
Educational development	MicroPython / Arduino / ESP-IDF teaching boards
USB devices	USB microphone, USB serial passthrough, USB MIDI keyboard, custom HID devices

6. Development Ecosystem

The ESP32-S3 has a mature development toolchain and community support:

Development method	IDE / toolchain	Language	Difficulty	Suitable for
ESP-IDF	VSCode + ESP-IDF extension	C	Intermediate-High	Commercial products, performance-sensitive applications
Arduino	Arduino IDE 2.x	C++	Intermediate	Rapid prototyping, Arduino ecosystem migration
MicroPython	Thonny IDE	Python	Low	Teaching, quick verification, scripted control

7. Relationship with This Tutorial

This tutorial is built around the **ESP32-S3 N16R8 core board** and covers the three mainstream development methods (ESP-IDF / Arduino / MicroPython), from basic GPIO peripherals to Wi-Fi network protocols, OTA upgrades, and AI voice interaction, organized into seven chapters in increasing difficulty:

Chapter	Content	Development method
1. Core board introduction	Chip overview, environment introduction, hardware details	—
2. General-purpose I/O	GPIO control / RGB LED / ADC / PWM	ESP-IDF / Arduino / MicroPython
3. Serial communication and storage systems	UART / I2C / SPI / NVS / LittleFS	ESP-IDF / Arduino / MicroPython
4. Network protocols and wireless communication	Wi-Fi scanning / TCP / UDP / MQTT / HTTP	ESP-IDF / Arduino / MicroPython
5. System and multimedia	Task scheduling / dual-core / interrupts / low power / audio playback and recording	ESP-IDF / Arduino / MicroPython
6. Advanced development	OTA upgrade	ESP-IDF / Arduino
7. XiaoZhi AI voice	Offline voice wake-up / LLM conversation / TTS	ESP-IDF